

Earthquake Search Prioritization in Urban Disaster Zones

1. Overview

Earthquakes are among the most destructive natural disasters due to their sudden onset and widespread structural damage. In urban environments, collapsed buildings, blocked roads, and disrupted communication systems create extreme challenges for search and rescue operations.

In the first **72 hours (“golden window”)**, survival probability drops rapidly. Therefore, efficient **search prioritization strategies** are critical to maximize rescue outcomes when resources such as drones, rescue teams, or thermal scanners are limited.

This document provides a structured approach for prioritizing search zones, optimizing limited resources, and guiding autonomous or semi-autonomous systems (e.g., drone swarms or AI agents) in disaster response scenarios.

2. Core Principles of Earthquake Search Prioritization

2.1 Time Sensitivity (Golden 72 Hours)

- Survival likelihood decreases significantly after 72 hours.
 - Priority must be given to **rapid-access zones** where survivors are most likely to be alive.
 - Early-stage missions should focus on **high-probability survival areas**, not full coverage.
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2.2 Population Density First

Areas with higher pre-disaster population density should be prioritized.

Heuristic:

- High-density residential buildings → High priority
 - Commercial hubs during working hours → Medium-High priority
 - Industrial or low-occupancy zones → Lower priority
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2.3 Structural Collapse Probability

Not all buildings collapse equally.

Priority increases when:

- Old infrastructure is present
- Poor seismic resistance design
- High-rise buildings with heavy vertical collapse risk

Lower priority:

- Open fields
 - Low-rise modern reinforced structures
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2.4 Accessibility and Reachability

Search prioritization must account for:

- Blocked roads
- Rubble density
- Drone flight accessibility

Zones that are:

- Easily reachable → scanned first (fast information gain)
 - Completely inaccessible → deferred or assigned aerial-only drones
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2.5 Communication Signal Presence

Even weak signals indicate possible survivors.

Priority increases if:

- Mobile signal spikes detected
 - Emergency SOS signals received
 - Thermal anomalies present
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3. Zonal Classification Strategy

Disaster areas are divided into structured priority zones:

Priority Zone 1 (Critical)

Characteristics:

- High population density
- Severe structural collapse
- Possible survivor signals detected

Action:

- Immediate drone deployment
 - Thermal scanning
 - Continuous monitoring
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● Priority Zone 2 (High)

Characteristics:

- Moderate population density
- Partial structural damage
- No confirmed survivor signals yet

Action:

- Scheduled scanning
 - Secondary drone allocation
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● Priority Zone 3 (Medium)

Characteristics:

- Low population density
- Light structural damage
- Accessible roads

Action:

- Batch scanning
 - Low battery drones assigned here
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● Priority Zone 4 (Low)

Characteristics:

- Open areas
- Industrial zones
- Minimal human presence

Action:

- Scanned only if resources allow